

Pratham Kwatra

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EDUCATION

Drexel University

June 2024

Bachelor of Science in Computer Science, Dean's List

Cumulative GPA: 3.86/4.00

Selected Coursework (G: Graduate Level, T: Taught the course as a TA):

Cloud Native Application Development (G), Data Structures and Algorithms (T), Operating Systems (T),

Web Development, Computer Networks, Software Design, Concurrency*

EXPERIENCE

Susquehanna International Group

March 2023 - Present

Monitoring Management Automation Development, Software Engineer Intern

Philadelphia, PA

- Engineered an **ASP.NET Core** and **Typescript** solution for real-time monitoring and alerting across 6000+ servers.
- Designed a **distributed service** to ingest ~1,250,000 entries per minute into a **partitioned Microsoft SQL Server** using the **Entity Framework**.
- Analyzed SQL execution plans and subsequently **optimized queries**, reducing deadlocks and improving ingestion speed by 300%.

Drexel University

September 2021 - Present

College of Computing and Informatics, Teaching Assistant

Philadelphia, PA

- Created a **Python** program to evaluate **free-form** and **source code** based solutions, speeding up the grading process by 20%.
- Prepared teaching material, assignments and graded 300+ students each time in **Operating Systems**, **Data Structures & Algorithms**, and **Introduction to Computer Science**.

Susquehanna International Group

March 2022 - September 2022

Middleware, Software Engineer Intern

Philadelphia, PA

- Designed a monitoring system using **Python** and the **Robot Framework** to track and warn about leftover processes created during runtime.
- Bolstered the overarching system reliability and performance by meticulously revitalizing and fine-tuning obsolete scripts in **Powershell**, **Bash**, and **Python**.
- Worked with **C++** team to optimize the design of a proprietary communication system to solve an **inconsistent deadlock** situation.

PROJECTS

RougeRobotRambo | *Python, PyGame*

- Created a **rouge-like game** called RougeRobotRambo by following the **AGILE development** method in a team of 4 people over three months.
- Worked with the team to create an AI enemy that would follow the player around and attack them using a **state machine** and **A* pathfinding algorithm**.

Voting System | *Go, Docker, Redis*

- Worked on creating a scalable voting system using **Go** and **Docker**.
- Used **Redis** to create a real-time system that would update the results of the election as soon as a vote was cast.

Online Sudoku | *React, JavaScript, Socket.IO, Postgres*

- Built a multiplayer Sudoku using **JavaScript** and **Socket.IO** in the backend.
- Designed a **Postgres** database that handled game scores, game status, and user information.

INVOLVEMENTS & ACHIEVEMENTS

- DragonHacks Hackathon 2020 - Best Community Award
- DragonHacks Hackathon 2021
- Drexel Algorithms & Data Structures Club
- Drexel CyberDragons (Cybersecurity Club)
- Drexel Weekend Warriors - Team Leader
- Drexel Badminton Club - Club Team